

are also more reliable and have higher operating limits (voltage and temperature). The memory chips then interface with a controller and a cache memory. The cache memory type depends on the main storage chips. MLC being cheaper and with greater density is found on entry-level and mid-range SSDs where space matters over speed. Either option is still faster than having a hard drive.

Cache

DRAM chips are used as cache in SSDs, they are pretty much the same as RAM and function in the same manner. Hard drives had cache memory which was limited to 64 MB and even that was too much. But SSDs have a much higher data transfer rate and thus need more cache memory. Typically a SSD has 256 MB to 1 GB of cache memory but there are entry-level models with cache memory as low as 32 MB.

Interface

SSDs use the same interface as HDDs except that the SSDs are more capable of utilising the full bandwidth that a SATA interface has to offer. The majority of SSDs in the market use the SATA3 interface which has a maximum data transfer limit of 6 Gb/s or 600 MB/s. There are more exotic SSDs which connect directly to the PCIE slot which has a much greater bandwidth. PCIE SSDs have a transfer rate of 830 MB/s, most SSDs lie in the 300 - 550 MB/s range.

Controller

This is at the core of the SSDs performance – The Controller. The controller's job entails keeping a check on the data stored in each chip and optimize the way data is to be stored in the SSD.

The flash chips used in SSDs have a limited number of write cycles. Therefore, a controller has to ensure that no one section is overused compared to others. The work done by a controllers includes deciding which block of data needs to be written and when to rewrite a block. Other functions include

- ▶ Error checking and correction (ECC)
- ▶ Keeping track of bad blocks of memory
- ▶ Read and write caching
- ▶ Garbage data collection (data stored in non-initialised regions)
- ▶ Encryption